

Fig. 6: One possible resulting configuration

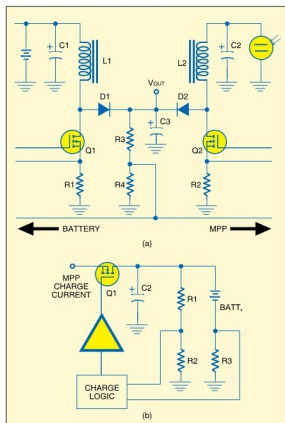


Fig. 7(a): Partial schematic of combined battery boost and MPP;
(b) partial schematic of simplified charger

ferent configurations until an optimum mix of cells per string and strings per MPP can be found that will maximize the efficiency of the solar cell portion of the system. (See Fig. 5.)

Once the solar-cell configuration is set, the next step is to perform a similar analysis on the battery system. We will be trading off the number of battery cells in each string, and the number of battery strings against the losses incurred in both the charging and boosting circuits. Fortunately, this is

actually a much simpler optimisation. We know the total number of battery cells from the storage requirements for the system. Therefore all we have to do is determine the optimum number of cells per string. As before, the easiest method to optimise this problem is a spreadsheet that calculates the overall loss due to the number of chargers and boost circuits.

We can start by making similar assumptions about increasing efficiency in the system. Shorter strings of battery cells can increase the efficiency of the charger/boost converters, by reducing the operating voltage. Shorter strings of battery cells can also increase the efficiency of the charger/boost converters by sharing the current requirements across

multiple converters. Longer strings of battery cells, which produce battery voltages near the bus voltage, can also increase the efficiency of the converters, by reducing the inductor current in the charger/boost converter. And, longer strings of battery cells also increase efficiency, by reducing the output current of the charger and input current of the boost converters. Each charger/boost converter will have a fixed loss dictated by its switching losses and the quiescent current of the converter.

The two main differences with the batteries is that we don't have a diode loss to confuse the issue, as we did with the solar cells, and there is a one-to-one relationship between the number of battery cell strings and the number of charger and booster pairs in the system. So, once we have the assorted loss equations figured out, we can plug the equations into a spreadsheet and vary the number of cells in each string until, again, we find the maximum efficiency configuration. (See Fig. 6.)

Have we finally squeezed every last percentage point out of the efficiency? Well, not quite. There are still a couple of things we can try, by thinking a little further outside the box:

1. We can redesign the battery boost switchers and the MPP converters to create the voltages required by the load, directly. This saves the losses, due to the load's switching regulators, at the cost of a little more complexity in the magnetics of the MPP and battery boost switchers. Fig. 7(a) shows the combined outputs of the two modules. Note the common output voltage feedback to both switching circuits.

2. We can combine the battery charger with the battery current output of the MPP modules. Basically, we're combining the switching logic of the charger with the current-bus control in the MPP converter blocks. Whenever a current pulse is routed to the batteries, instead of storing it in a capacitor and then re-chopping the current in the charger, we just use the pulsed current from the MPP directly, by switching it to those batteries that require charging. It saves the second set of switching losses and magnetic losses in the load's switching regulator, and any capacitor ESR losses incurred in storing the pulse prior to re-chopping. (See Fig. 7(b).)

3. Finally, we can eliminate losses incurred during the slow incremental dither in the MPP, by replacing it with an intelligent dither that uses a form of successive approximation to more quickly find the maximum power point. Further, the intelligent dither can avoid any periodic losses due to a